H E R B S E N S E

AI-Based Herbal Plant Identification and Digital Herbarium System



Presents by Group 38

Introduction

Medicinal plants play an important role in traditional healthcare, but identifying them correctly from appearance alone can be difficult, especially when similar-looking plants are involved. HerbSense is an AI-based web application designed to identify medicinal plants from photographs using a YOLO11n deep-learning model. When the system is uncertain, it asks users additional questions about plant features and combines this information with the image prediction. It also provides digital herbarium information, look-alike comparisons, and safety warnings to help users make more informed and cautious decisions.

Problem & Objectives

- Identifying medicinal plants from appearance alone can be difficult, especially with look-alike species.
- Existing photo-based tools may provide a single prediction without enough support when confidence is low.

OUR GOAL

build a guided, safety-aware system that identifies plants and shows how trustworthy the result is.

MAIN OBJECTIVES

- Detect and classify medicinal plants from a photo
- Use confidence levels and user-provided plant features to verify uncertain predictions.
- Provide herbarium information, look-alike comparisons and safety warnings
- Maintain a secure web application with accounts and scan history.

Our AI Solution - Hybrid AI

01

YOLO11N DEEP LEARNING MODEL

detects the plant and returns candidate classes with confidence scores.

02

CONFIDENCE POLICY

decides whether the result is high, medium or low confidence.

03

RULE-BASED EXPERT SYSTEM

asks feature questions such as leaf arrangement, leaf shape, scent and growth habit.

Our AI Solution - Hybrid AI

04

WEIGHTED EVIDENCE FUSION

Final score = 0.6 × image score + 0.4 × rule score

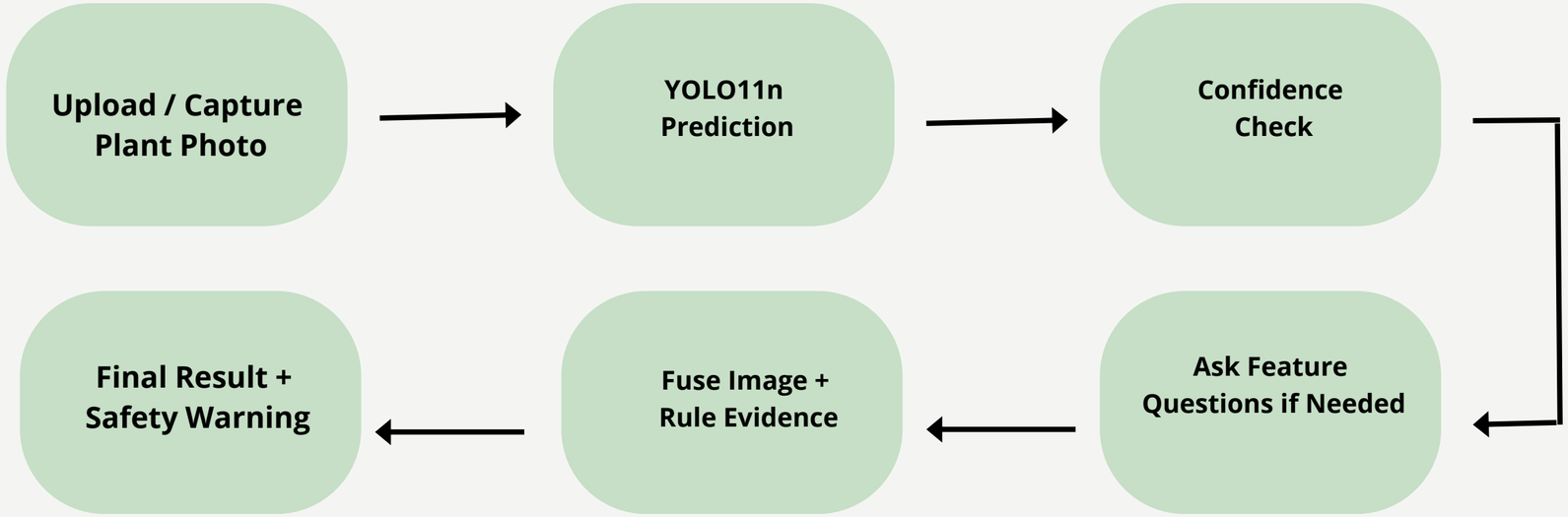
02

LANGUAGE MODEL

generates a short description only; it does NOT influence plant identification.

This combination makes the system more explainable and safer than relying only on an image prediction.

SYSTEM WORKFLOW



- High ≥ 75% → result shown immediately.
- Medium 45–74% → verification questions + caution.
- Low < 45% → verification questions + critical warning

System Features & Technology

01



Digital Herbarium

information for 12 medicinal plants, including uses, habitat, precautions and feature values.

02



Look-alike comparison to help users understand similar species

03



Severity-graded safety warnings based on confidence.

04



User accounts and scan history.

Technology stack:

01 **Frontend**

React 18, Vite, Tailwind CSS

02 **Backend**

Node.js + Express

03 **Database**

PostgreSQL

04 **AI**

YOLO11n via Roboflow + rule-based expert system + optional LLM

05 **Security**

server-side API keys, bcrypt, JWT, input validation and restricted CORS

Testing & Results

Automated testing

15 out of 15 tests passed.

Verification simulation

132 ordered plant-pair cases were evaluated.

When the wrong plant was initially ranked first :

- **All four correct feature answers → 100.0% corrected**
- **Leaf arrangement + scent → 97.0% corrected**
- **Leaf shape + growth habit → 90.9% corrected**
- **Scent only → 80.3% corrected**

These results demonstrate that the verification logic can correct many wrong first predictions under idealised conditions.

Limitations & Future Work

01

02

03

04

Limitations

- Image-model accuracy has not yet been independently evaluated.
- Herbarium is limited to 12 plants and four main features.
- Confidence thresholds and fusion weights are design choices and not calibrated on held-out data.

Future Works

- Evaluate and calibrate the model using independent field photos.
- Expand the herbarium with reliable sources and more look-alikes.
- Add Sinhala and Tamil interfaces and improve grounding of LLM description

Conclusion

HerbSense combines deep learning, rule-based reasoning, and safety-aware feedback to make medicinal plant identification more transparent and cautious. The system can identify plants from images, verify uncertain predictions using additional plant features, and provide herbarium information, look-alike comparisons, and safety warnings. The results show that the verification layer can effectively correct many wrong predictions under idealised conditions. Overall, HerbSense demonstrates how different AI techniques can work together to provide a more reliable and user-friendly plant identification system.

T H A N K Y O U